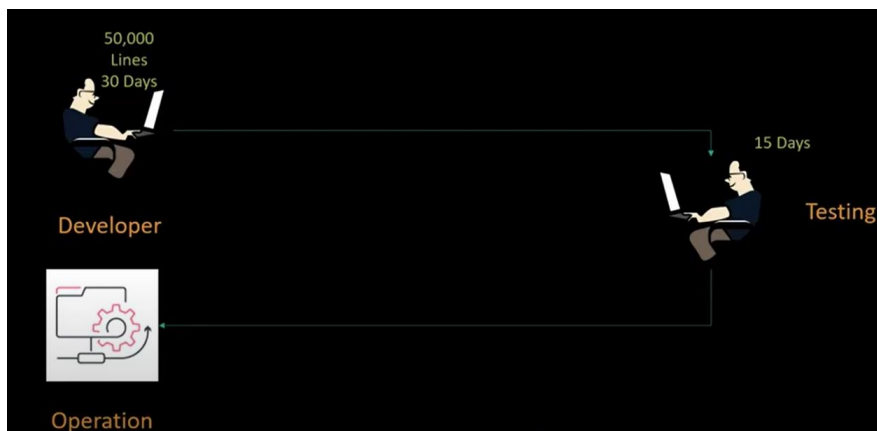


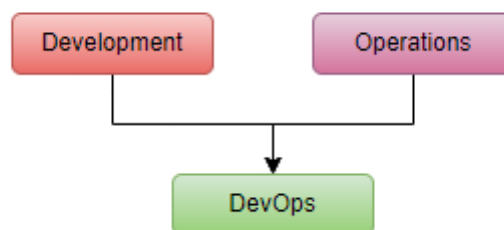
AWS DevOps

Old Days:



DevOps: Is a methodology but not the tools DevOps makes the S/W development and release very much easy and fast.

Continuous Build and Automate product Delivery



Continuous Deployment/Delivery

Starts after build generates artifacts

Continuous Integration with source code-> artifact

Continuous Deployment, application deployed on the servers and available to end-users

Continuous Delivery (CI/CD is continuous delivery to business)

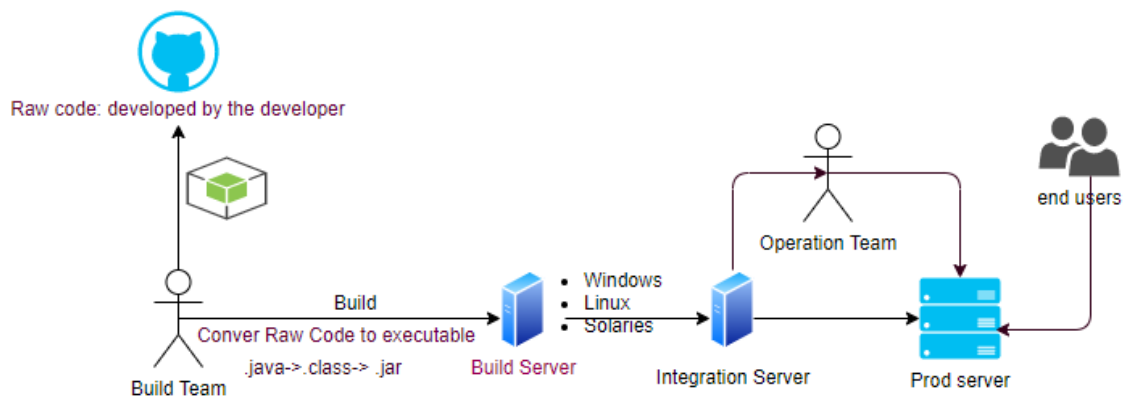
UC: We have developed web application have 50 functionalities, packaged it in artifacts (.war/. ear) and expected to support 1000 concurrent users.

step1: write code

step2: build it, deploy on integration server, can support 1000 concurrent user? Are all 50 functionalities running fine?

step3: if desired behavior is ok, then deploy it on production.

step4: monitor, if there are 2000 concurrent users then behavior?



DevOps is a process to automate the all steps using proper tools.

statement: application running fine in dev machine, but not running in qa machine?

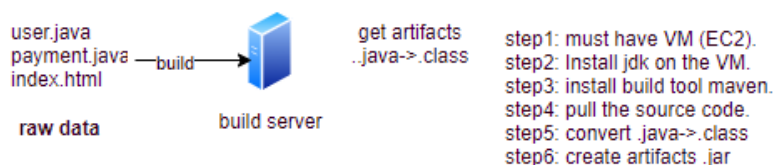
measure: is all configuration available? Is version in both environments are same?

action: package all the dependencies together and deploy, automate the process.

result: less time of deployment, less chance of failure.

UC: If instead of package, someone give us Raw Source Code?

First, we have to build the Raw Source Code.



Cloud means everything is virtual.

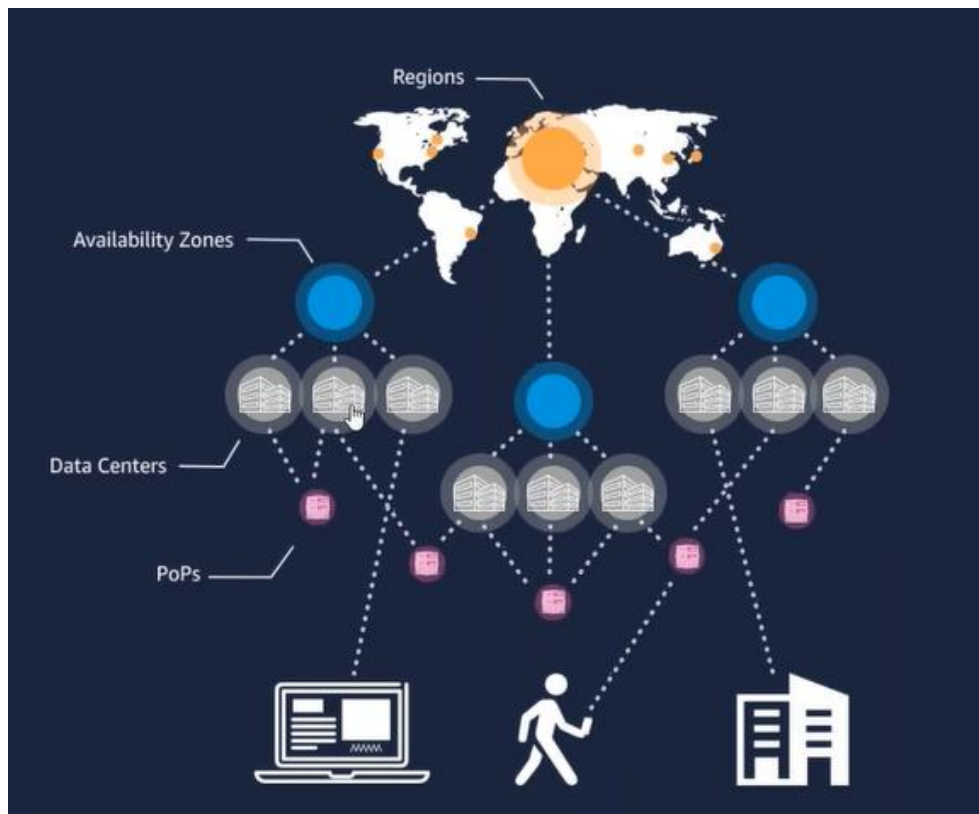
step1: Infrastructure in the cloud.

step2: Application deployment over Cloud IT resources.

Infrastructure and Application Deployment



AWS Core Services



- Regions
- AZ
- Data Centers
- POP (reduce latency collaborate with third party): Using DNS and anycast (reduce latency nearest POP)

status.aws.amazon.com

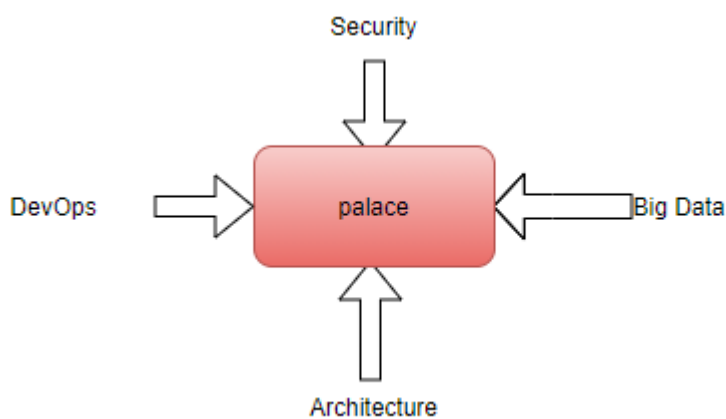
#Click on personal health dashboard (**global service**): service used in your account (outage).

Event log # within the personal health dashboard

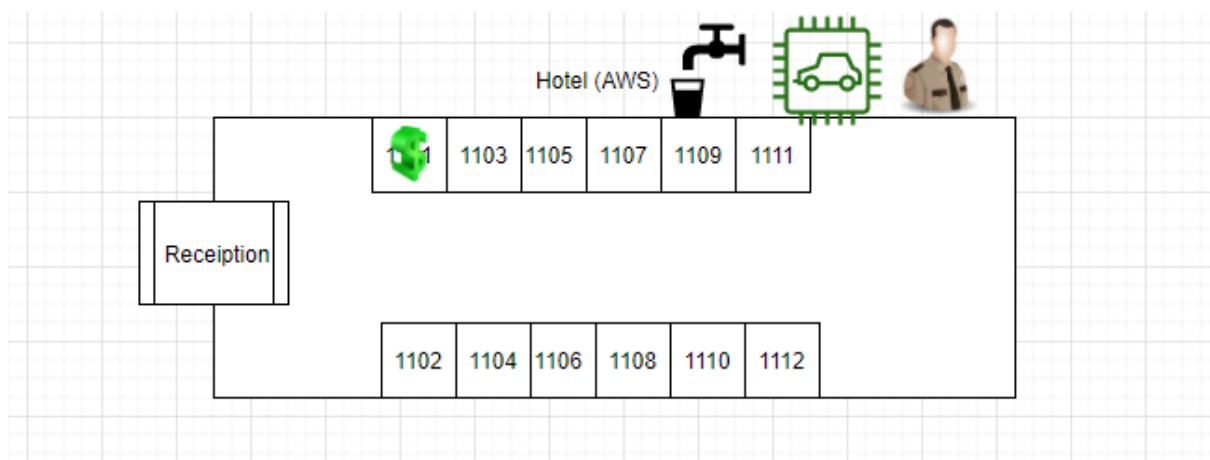
cloudharmony.com # different-different cloud service provider

Cloud Concept for each and every vendor almost same, just like a car manufactured by different vendors.

Rule for Palace (must be follow, you enter from any side)

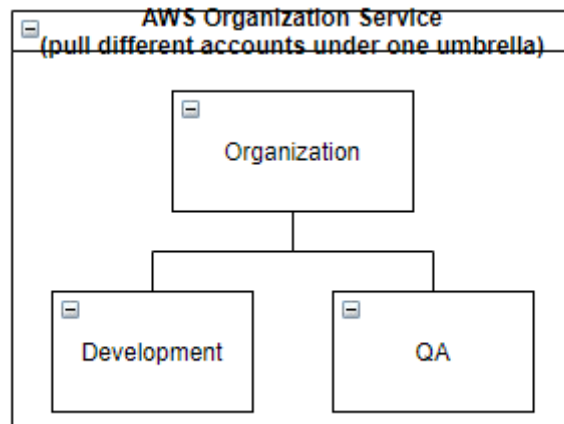


Shared Security Responsibility Model:

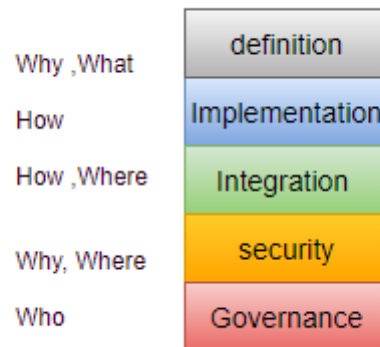


AWS Organization Account:

Define a service policy at organization level will by default applicable to all child accounts.



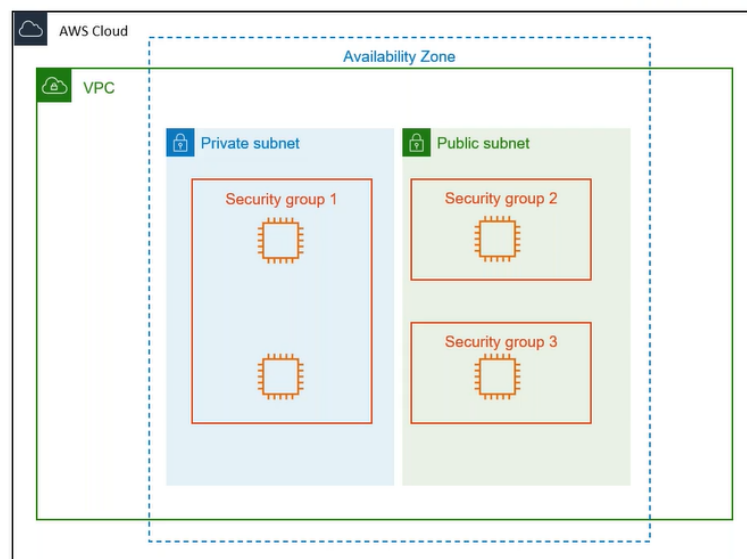
Each service comes with different layers



Using different-different services we automate our process; DevOps enables the same.

DevOps is about creating an environment and then deploy the application, after that making them secure and monitoring.

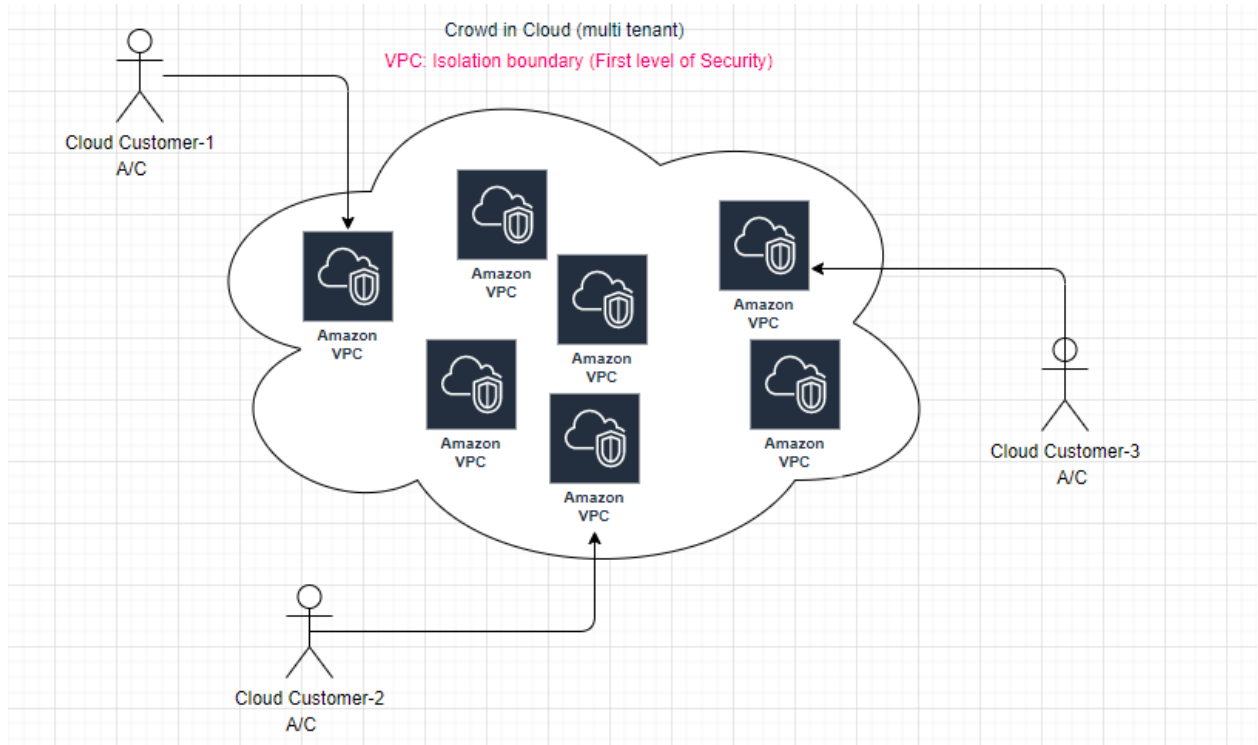
- VPC
 - Subnets
 - Internet Gateway
 - NACL
- Subnets
 - Private
 - Public
 - Route table
- IAM
 - User
 - Groups
 - Policies
 - Role
- Launching an EC2
 - Instance Types
 - Pricing Models
 - Vertical Scaling
 - Elastic IP
 - Security Group



Setup:

How I Isolate my resources from other's resources?

VPC:



VPC- one default VPC per region

Subnet- subnet corresponding to each AZ.

public subnet: associated with IGW (entry of igw in associated route table),
auto assign public ip

private subnet: not associated with IGW (NAT Gateway)

main objective of VPC is to isolate our resources , never use the pre-configured VPC (spans over all AZ with in the region).

IP address range:

With Amazon BYOIP you can bring part, or all your public IPv4 address range from your on-premises deployment to your AWS account. You continue to own the address range, but AWS advertises it on the internet on your behalf.

When using Amazon BYOIP there are several steps involved in bringing your IP address range to AWS, including preparing your range, provisioning, advertising, and allocating Elastic IP addresses.

[Good blog related to BYOIP](#)

Private address range		
Class	start address	finish address
A	10.0.0.0	10.255.255.255
B	172.16.0.0	172.31.255.255
C	192.168.0.0	192.168.255.255

Public address range		
Class	start address	finish address
A	0.0.0.0	126.255.255.255
B	128.0.0.0	191.255.255.255
C	192.0.0.0	223.255.255.255
D	224.0.0.0	239.255.255.255
E	240.0.0.0	254.255.255.255

private address don't move between different networks.

step1: create your own VPC (LAN-default 5 vpc per account): private IP range

Create VPC Info

A VPC is an isolated portion of the AWS cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block Info

IPv6 CIDR block Info

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy Info

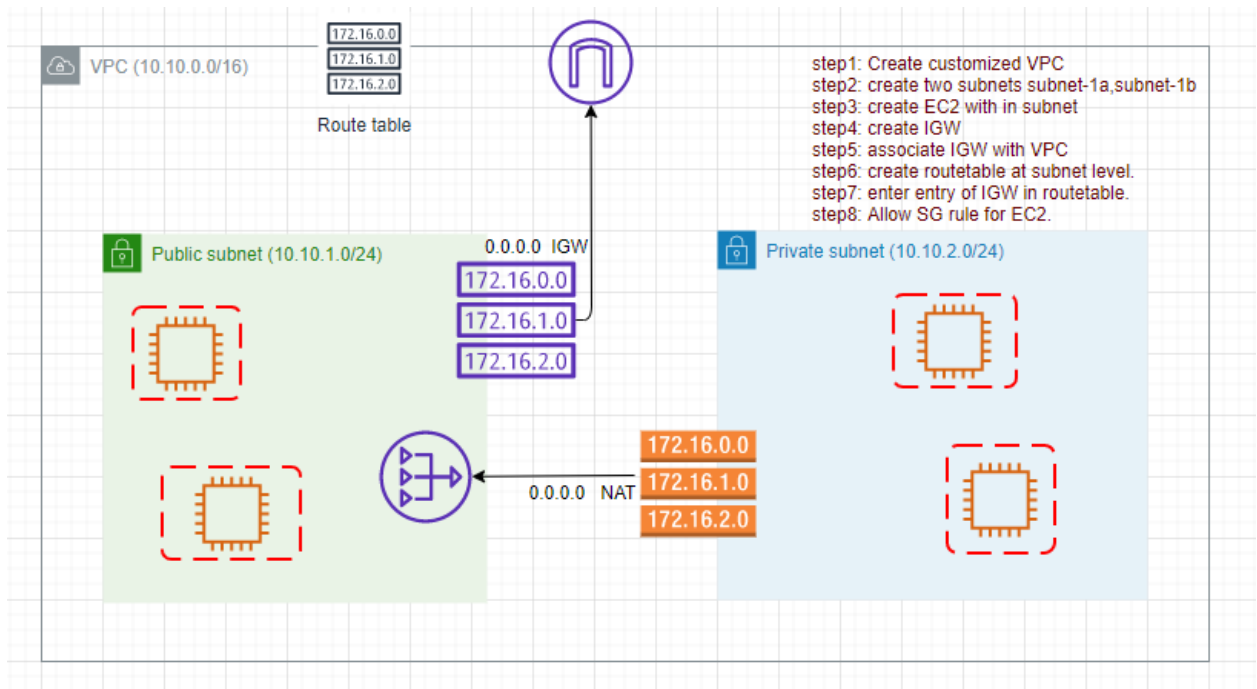
Default

step2: By default one subnet per AZ is available, in customized vpc create own two subnets in AZ1 and AZ2, and create EC2 instances with in the subnets.

step3: As of now EC2 is not accessable to the Internet.

step4: Create your own IGW (attach with VPC).

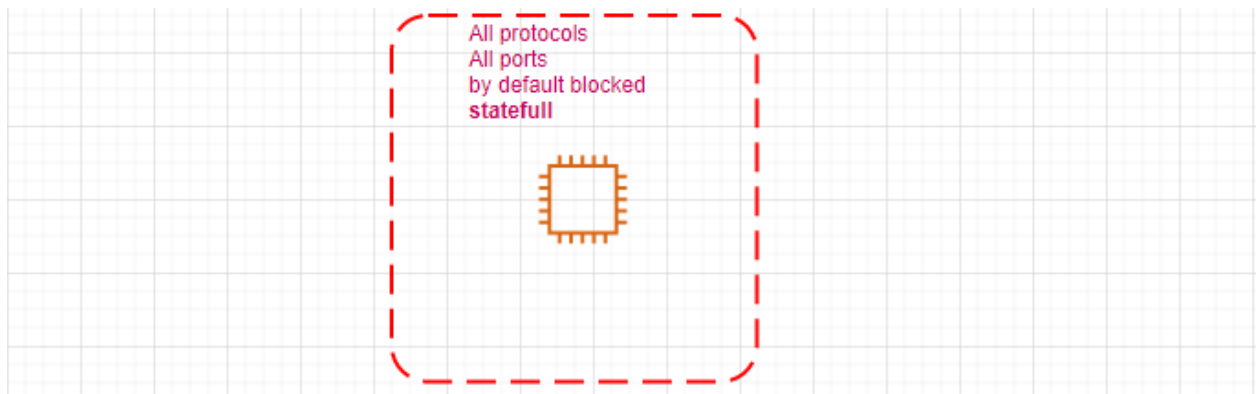
step5: Create route table associated with subnet.



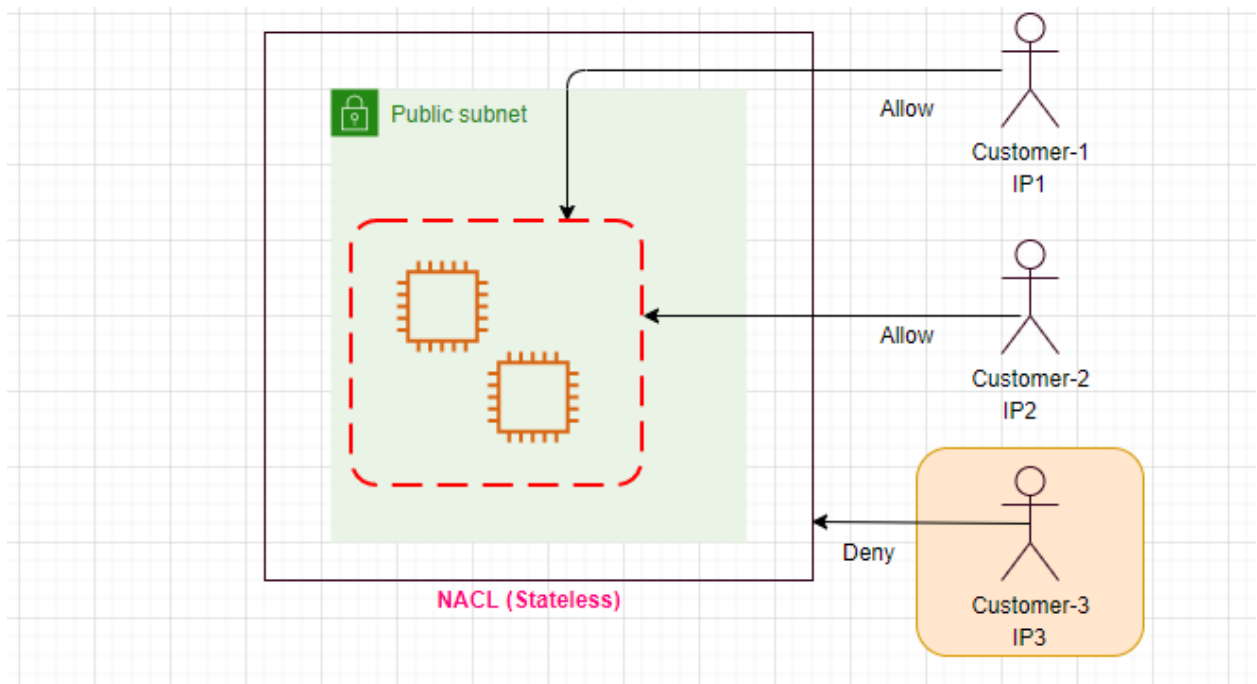
NAT Gateway: EC2 instance inside private subnet access internet through the NAT.

DHCP issues private IP for EC2 .

SG (EC2 level): Deny all traffic by default (Statefull).

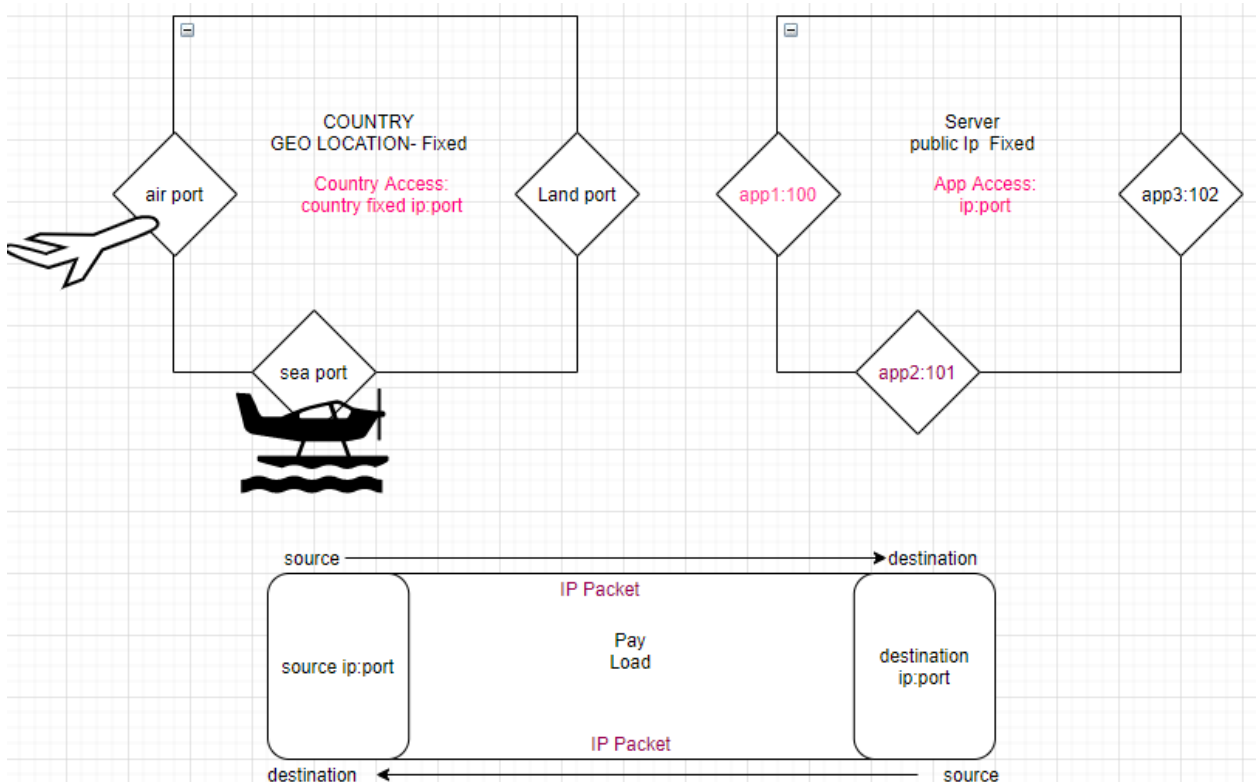


NACL (subnet level): Allows all traffic by default (Stateless). Rule are evaluated sequentially.



ports?

entry point, without port can't run more than one application on the same server.



BYOIP: Bring your own IP, all public IP are paid, by default AWS assigns public ip address from its owned pool of public ip addresses.

Elastic IP: static IP address (associate/release)

status check:

system status check: Infrastructure in which EC2 is hosting is correct

instance status check: Is that Instance reachable?

public subnet: - EC2 instances have public as well as private IP addresses.

private subnet: - EC2 instances have only private IP addresses.